A

Major Project

On

A CENTRALIZED STUDENT–FACULTY PLATFORM FOR REAL-TIME ACADEMIC PERFORMANCE PREDICTION

(Submitted in partial fulfilment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

M Anjali	(227R1A1236)
G Abhinav	(237R5A1204)
M Akhila Kumari	(227R1A1238)

Under the Guidance of

D Mahesh Babu

(Assistant Professor)

DEPARTMENT OF INFORMATION TECHNOLOGY

CMR TECHNICAL CAMPUS

UGC AUTONOMOUS

(Accredited by NAAC, NBA, Permanently Affiliated to JNTUH, Approved by AICTE, New Delhi)

Recognized Under Section 2(f) & 12(B) of the UGCAct.1956,

Kandlakoya (V), Medchal Road, Hyderabad-501401.

May, 2026.

DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the project entitled "**A CENTRALIZED STUDENT–FACULTY PLATFORM FOR REAL-TIME ACADEMIC PERFORMANCE PREDICTION**" being submitted by **M Anjali (227R1A1236), G Abhinav (237R5A1204) & M Akhila Kumari (227R1A1238)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

D Mahesh Babu
Assistant Professor
INTERNAL GUIDE

HoD

Dr. A. Raji Reddy
DIRECTOR

Signature of External Examiner

Submitted for viva voice Examination held on _____

ACKNOWLEDGEMENT

We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project, we take this opportunity to express our profound gratitude and deep regard to our guide **D Mahesh Babu,** Assistant Professor for his exemplary guidance, monitoring and constant encouragement throughout the project work. The blessing, help and guidance given by him shall carry us a long way in the journey of life on which we are about to embark.

We take this opportunity to extend our heartfelt appreciation to the Project Review Committee (PRC) Coordinators— **Dr. J. Narasimharao, Mr. Md. Sajid Pasha, Mr. K. Praveen Kumar, Mr. D. Krishna Kumar, Mr. Y. Satyam and Mr. K. Srinu** —for their unwavering support, insightful guidance, and valuable inputs, which played a crucial role in steering this project through its various stages.

Our sincere appreciation also goes to our **Head of the Department**, for encouragement and continuous support in ensuring the successful completion of our project.

We are deeply grateful to **Dr. A. Raji Reddy**, Director, for his cooperation throughout the course of this project. Additionally, we extend our profound gratitude to Sri. **Ch. Gopal Reddy**, Chairman, Smt. **C. Vasantha Latha**, Secretary and Sri. **C. Abhinav Reddy**, Vice-Chairman, for fostering an excellent infrastructure and a conducive learning environment that greatly contributed to our progress.

We also acknowledge and appreciate the guidance and assistance provided by the faculty and staff of **CMR Technical Campus**, whose contributions have been invaluable in bringing this project to fruition.

Lastly, we sincerely thank our families for their unwavering support and encouragement. We also extend our gratitude to the teaching and non-teaching staff of CMR Technical Campus for their guidance and assistance. Their contributions, along with the support of everyone who helped directly or indirectly, have been invaluable in the successful completion of this project.

M Anjali (227R1A1236)
G Abhinav (237R5A1204)
M Akhila Kumari (227R1A1238)

VISION AND MISSION

INSTITUTE VISION:

To Impart quality education in serene atmosphere thus strive for excellence in Technology and Research.

INSTITUTE MISSION:

1. To create state of art facilities for effective Teaching- Learning Process.

2. Pursue and Disseminate Knowledge based research to meet the needs of Industry & Society.

3. Infuse Professional, Ethical and Societal values among Learning Community.

DEPARTMENT VISION:

To produce technically proficient and skilled graduates in Information Technology and to prepare them to meet the needs of industry and society in a holistic learning environment.

DEPARTMENT MISSION:

1. To prepare students with the essential knowledge and contemporary abilities for solving emerging technological problems.

2. To train the students on cutting – edge computing facilities with industry interaction to enhance in practical exposure to real world situations.

3. To enhance the leadership qualities and team spirit with social values to serve the society.

ABSTRACT

The accurate monitoring and prediction of student performance is essential for enhancing learning outcomes and enabling data-driven academic decisions. Traditional academic management systems primarily function as static repositories for grades and attendance records, lacking predictive analytics capabilities. Moreover, such systems often do not provide integrated communication channels among administrators, faculty, students, and parents, limiting their effectiveness in improving student engagement and support. This work presents a centralized student portal that leverages machine learning (ML) algorithms to analyze historical academic data and predict individual student performance. Both models are trained on past student marks, and performance is evaluated using accuracy, precision, recall, and F1-score. Experimental results show that XGBoost outperforms Naive Bayes, achieving an accuracy exceeding 93% in predicting performance categories such as Good, Fair, or Poor. These predictions provide early indicators for intervention and personalized academic guidance. The system incorporates three dashboards such as Admin, Faculty, and Student with distinct role-based functionalities. The Admin Dashboard allows the addition and management of faculty and student records, viewing of institutional performance, and monitoring of teacher effectiveness based on aggregated student grades. The Faculty Dashboard supports attendance management, assignment creation, study material uploads, mark entry, student communication via email notifications, and viewing of ML-generated performance predictions. The Student Dashboard enables access to assignments, downloading of study materials, marks visualization with predictive feedback, and direct messaging to faculty. Additional features include a faculty suggestion module, enabling students to choose instructors based on algorithmically determined performance ratings or manual selection from feedback and reviews. The system also implements faculty capacity constraints (e.g., 60 students per faculty) and facilitates structured feedback collection from students. The proposed system provides an integrated platform for academic tracking, communication, and performance prediction. By embedding ML-based predictive analytics within a centralized portal, the proposed framework supports proactive interventions, fosters collaboration among stakeholders, and enhances transparency in academic performance management..

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
Figure 3.1	Project Architecture of the Centralized Student-Faculty Platform for Real-Time Academic Performance Prediction	13
Figure 3.2	Class Diagram	16
Figure 3.3	Use Case Diagram	17
Figure 3.4	Sequence Diagram	18
Figure 3.5	Component Diagram	18
Figure 3.6	Deployment Diagram	19
Figure 3.7	Activity Diagram	19
Figure 3.8	Dataflow Diagram	21
Figure 4.1	Internal architecture of Naïve Bayes classifier for student performance prediction.	23
Figure 4.2	Flow diagram of XGBoost classifier training process.	25

FIGURE NO	FIGURE NAME	PAGE NO
Figure 5.1	Interface Overview	37
Figure 5.2	Uploading Teacher' Details	38
Figure 5.3	Uploading Student Details	38
Figure 5.4	Teacher Uploading Student Attendance	39
Figure 5.5	Teacher Creating Assignments	39
Figure 5.6	Student's Checking Assignments	40
Figure 5.7	Student Downloading Learning Materials	40

LIST OF TABLES

TABLE NO	TABLE NAME	PAGE NO
Table 6.2.1	Comparative Analysis	42
Table 6.2.2	Performance Metrics of Navy Bayes Classifier	43
Table 6.2.3	Performance Metrics of XG Boost Classifier	43

TABLE OF CONTENTS

ABSTRACT i
LIST OF FIGURES ii
LIST OF TABLES iv
1. INTRODUCTION 1
1.1 PROJECT PURPOSE 1
1.2 PROJECT FEATURES 2
2. LITERATURE SURVEY 4
2.1 REVIEW OF RELATED WORK 6
2.2 DEFINITION OF PROBLEM STATEMENT 8
2.3 EXISTING SYSTEM 8
2.4 PROPOSED SYSTEM 10
2.5 OBJECTIVES 11
2.6 HARDWARE & SOFTWARE REQUIREMENTS 11
2.6.1 HARDWARE REQUIREMENTS 11
2.6.2 SOFTWARE REQUIREMENTS 12
3. SYSTEM ARCHITECTURE & DESIGN 13
3.1 PROJECT ARCHITECTURE 13
3.2 DESCRIPTION 14
3.3 UML DIAGRAMS 15
3.3.1 CLASS DIAGRAM 15
3.3.2 USE CASE DIAGRAM 16
3.3.3 SEQUENCE DIAGRAM 17
3.3.4 COMPONENT DIAGRAM 18
3.3.5 DEPLOYMENT DIAGRAM 19
3.3.6 ACTIVITY DIAGRAM 19
3.3.7 DATA FLOW DIAGRAM 20
4. IMPLEMENTATION 22
4.1 ALGORITHMS USED 22
4.2 SAMPLE CODE 28
5. RESULTS & DISCUSSION 37
6. VALIDATION 41
6.1 INTRODUCTION 41
6.2 TEST CASES 42
6.2.1 COMPARATIVE ANALYSIS 42
6.2.2 PERFORMANCE METRICS OF NAÏVE BAYES CLASSIFIER 43
6.2.3 PERFORMANCE METRICS OF XGBOOST CLASSIFIER 43

7. CONCLUSION & FUTURE ASPECTS 44
7.1 PROJECT CONCLUSION 44
7.2 FUTURE ASPECTS 45
8. BIBLIOGRAPHY 46
8.1 REFERENCES 46
8.2 GITHUB LINK 48

1. INTRODUCTION

1. INTRODUCTION

The project titled **"A Centralized Student–Faculty Platform for Real-Time Academic Performance Prediction"** focuses on developing an intelligent system that enables efficient monitoring and prediction of student academic performance. The platform integrates academic data such as marks, attendance, assignments, and faculty feedback into a single centralized portal. Using machine learning algorithms, the system analyzes historical student data to identify performance patterns and predict future academic outcomes. These predictions help classify students into categories such as Good, Fair, or Poor, allowing educators to better understand student progress and learning trends.

Since academic institutions generate large volumes of student data, manual monitoring and analysis become inefficient and time-consuming. Therefore, an automated machine learning–based system is necessary to analyze academic records and provide meaningful insights in real time. The proposed system improves communication between administrators, faculty, and students while providing dashboards for managing academic activities and performance analytics. By enabling early identification of students who require additional academic support, the system helps institutions take proactive actions, improve learning outcomes, and enhance transparency in academic performance management.

1.1 PROJECT PURPOSE

The primary purpose of this project is to enhance academic performance monitoring in educational institutions through a centralized student–faculty platform. Traditional academic management systems mainly store information such as marks and attendance but do not provide intelligent insights about student progress. Because of this limitation, faculty members often find it difficult to identify students who may require additional academic guidance at an early stage. The proposed system integrates multiple academic activities such as assignments, attendance, and marks into a single platform. By applying machine learning techniques to historical academic data, the system predicts student performance categories and provides meaningful insights to educators and students.

With the growing amount of academic data generated in institutions, manual monitoring and analysis become time-consuming and inefficient. Therefore, an automated and intelligent system is required to analyze data and provide timely feedback. The proposed platform helps faculty identify at-risk students and enables early academic intervention. It also improves communication between students and faculty through an integrated portal. By reducing manual effort and improving decision-making through predictive analytics, the system contributes to better academic outcomes and a more transparent and efficient learning environment.

1.2 PROJECT FEATURES

This project incorporates several key features to improve the efficiency of academic management and student performance prediction within educational institutions:

Centralized Academic Management:

The system provides a centralized platform where academic information such as attendance, assignments, marks, and faculty feedback are stored and managed in a single database. This eliminates the need for fragmented record-keeping systems and allows administrators, faculty, and students to access academic information in a structured and organized manner. The centralized database ensures better data consistency, transparency, and ease of monitoring student progress.

Machine Learning-Based Performance Prediction:

The project integrates machine learning algorithms such as Naïve Bayes and XGBoost to analyze historical academic data and predict student performance. These models evaluate factors such as marks, course details, and academic history to classify student performance into categories like Good, Fair, or Poor. This predictive capability helps faculty identify students who may need additional support and enables timely academic intervention.

Interactive Dashboards and Communication Features:

The system includes separate dashboards for Admin, Faculty, and Students, each providing role-based functionalities. Faculty members can upload study materials, manage attendance, assign coursework, and communicate with students through messages or email notifications. Students can view assignments, download learning materials, check academic reports, and receive machine learning-based feedback on their performance. This improves communication, enhances transparency, and supports better academic engagement.

By integrating these features, the proposed system provides an efficient, intelligent, and scalable solution for academic performance monitoring and student–faculty interaction, helping institutions improve academic decision-making and overall learning outcomes.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

With the rapid growth of digital learning platforms and online academic management systems, monitoring student performance has become an essential requirement for educational institutions. Universities and colleges generate large volumes of academic data such as attendance records, assignment submissions, examination scores, and faculty feedback. Although many institutions maintain digital records, these systems often function only as data storage tools and lack intelligent analytics for predicting student performance. This limitation highlights the need for advanced machine learning techniques capable of analyzing academic data and identifying patterns that influence student success.

This literature survey reviews previous research on student performance prediction and academic management systems. It analyzes existing techniques including traditional statistical analysis methods, machine learning models, deep learning approaches, and educational data mining techniques. These methods aim to improve prediction accuracy and support early identification of students who may require additional academic guidance.

Early research in academic performance prediction relied on traditional statistical techniques and simple data analysis methods. Educational institutions primarily used spreadsheets and basic database systems to record academic data such as marks and attendance. These systems helped store information but did not provide predictive insights. Researchers later began exploring statistical models that used historical academic data to identify relationships between learning behaviors and performance outcomes. However, these models had limited ability to handle complex datasets and could not accurately predict future academic performance.

With the development of machine learning, researchers started applying algorithms such as Decision Trees, Support Vector Machines (SVM), Random Forest, and Logistic Regression to educational datasets. These models analyzed various academic features including attendance rates, assignment scores, demographic information, and course performance. Studies demonstrated that machine learning models could effectively identify patterns associated with student success or failure. Random Forest models, in particular, were found to provide higher prediction accuracy due to their ability to analyze multiple decision trees simultaneously.

Educational Data Mining (EDM) also emerged as an important research field focused on extracting useful insights from learning data. Researchers analyzed datasets from Learning

Management Systems (LMS) such as Moodle to study student engagement patterns and academic progress. By applying clustering and classification algorithms, they identified correlations between course activities and final grades. These studies showed that student interaction with learning resources plays a significant role in academic success.

Recent research has explored the use of deep learning techniques to improve prediction accuracy. Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and hybrid deep learning models have been used to analyze large academic datasets. These models are capable of capturing complex nonlinear relationships between multiple academic variables. Although deep learning models often achieve higher prediction accuracy, they require large datasets and higher computational resources compared to traditional machine learning algorithms.

Another important advancement in educational analytics is the use of multidimensional datasets. Researchers have started combining multiple factors such as student demographics, academic history, behavioral patterns, and institutional indicators to build more accurate prediction models. By analyzing these factors together, machine learning algorithms can better understand the learning environment and identify students who may be at risk of poor academic performance.

Researchers have also explored the use of ensemble learning techniques to improve model stability and accuracy. Ensemble models combine predictions from multiple algorithms to reduce overfitting and improve reliability. For example, integrating models such as Random Forest with boosting techniques has shown improved performance in predicting academic outcomes. Ensemble approaches help create more robust predictive systems capable of handling diverse academic datasets.

Another important development in student performance prediction research is the integration of predictive models within academic management platforms. Instead of using standalone prediction systems, researchers have proposed centralized portals where machine learning models operate alongside academic data management tools. Such platforms allow faculty members and administrators to monitor student progress in real time and provide early interventions when necessary.

Transfer learning and automated feature selection techniques have also been investigated to enhance prediction performance. These techniques reduce the need for manual feature engineering and enable models to learn from existing datasets efficiently. By using optimized feature selection methods, researchers can identify the most significant factors

affecting student performance.

Despite these advancements, several challenges remain in developing reliable academic performance prediction systems. Traditional data management systems lack predictive capabilities, while many machine learning models require large amounts of clean data and careful preprocessing. Additionally, most existing systems focus only on prediction without integrating communication and academic management functionalities.

To address these limitations, the proposed system integrates centralized academic data management with machine learning-based performance prediction. By combining machine learning algorithms with an interactive student–faculty platform, the system enables real-time monitoring of academic performance, improves communication between stakeholders, and supports proactive academic interventions.

Overall, the literature survey highlights the evolution of student performance prediction techniques from basic statistical analysis to advanced machine learning and deep learning models. The integration of predictive analytics with centralized academic management platforms provides a promising approach for improving learning outcomes and institutional decision-making.

2.1 REVIEW OF RELATED WORK

The prediction and monitoring of student academic performance have been widely studied in the fields of educational data mining, machine learning, and learning analytics. Various approaches have been proposed to analyze academic datasets and identify factors that influence student success. This section reviews existing research methodologies and highlights their strengths and limitations.

1. Traditional Academic Monitoring Approaches

Early academic monitoring systems relied on manual record keeping and basic database management systems. Educational institutions maintained student information such as attendance, examination scores, and assignments using spreadsheets or physical registers. These systems primarily served as storage tools and lacked analytical capabilities. As a result, identifying students who required academic support depended entirely on manual evaluation by faculty members.

2. Machine Learning-Based Approaches

With advancements in machine learning, researchers began applying classification algorithms such as Support Vector Machines (SVM), Decision Trees, Random Forest, and Logistic Regression to academic datasets. These algorithms analyze multiple academic indicators including marks, attendance, course engagement, and demographic data. Machine learning models demonstrated improved prediction accuracy compared to traditional statistical methods, enabling early identification of students who may be at risk of poor academic performance.

3. Educational Data Mining Approaches

Educational Data Mining techniques have also been widely used to analyze learning patterns and student behavior. Researchers utilize data from Learning Management Systems to examine interactions between students and digital learning resources. By applying clustering and classification algorithms, these studies reveal relationships between learning engagement and academic performance.

4. Recent Advances: Deep Learning and Hybrid Models

Recent studies have explored the use of deep learning architectures such as Artificial Neural Networks and hybrid machine learning models for academic performance prediction. These models can capture complex relationships between multiple academic factors and provide improved prediction accuracy. However, deep learning models often require large datasets and higher computational resources.

5. Comparison with the Proposed Approach

Although existing studies have demonstrated the effectiveness of machine learning techniques in predicting student performance, many systems focus only on predictive modeling without integrating academic management functionalities. The proposed system extends previous research by combining predictive analytics with a centralized student–faculty platform. By integrating machine learning models such as Naïve Bayes and XGBoost with role-based dashboards, the system improves academic monitoring, communication, and decision-making in educational institutions.

2.2 DEFINITION OF PROBLEM STATEMENT

The primary goal of this project is to develop a centralized and intelligent system capable of managing academic activities while predicting student performance using machine learning techniques. The system integrates academic data such as attendance, assignment submissions, examination marks, and faculty feedback into a single unified platform to enable effective monitoring and analysis of student progress. Many educational institutions currently rely on fragmented or manual systems that store academic data in separate locations, making it difficult for faculty and administrators to track student performance and identify students who require additional support. By applying machine learning algorithms such as Naïve Bayes and XGBoost to historical academic data, the proposed system aims to generate accurate performance predictions, support early identification of academically at-risk students, and improve communication between students and faculty to enhance overall educational outcomes.

2.3 EXISTING SYSTEM

In many academic institutions, student performance management and communication between faculty and students are handled using traditional or semi-digital systems. Academic data such as marks, attendance, and assignments are often recorded in registers, spreadsheets, or separate software applications that are not interconnected. Faculty members maintain attendance records manually, while assignment distribution and academic communication frequently occur through informal platforms such as email or messaging applications.

These existing systems lack centralized data integration, making it difficult to obtain a comprehensive view of student academic progress. Since academic records are stored in multiple locations, compiling data for performance analysis becomes time-consuming and prone to human error. Faculty members must manually review marks and attendance records to identify students who may require additional support.

Furthermore, traditional academic management systems do not include predictive analytics capabilities. They simply store academic data without analyzing it to generate insights about future performance trends. As a result, institutions often fail to identify academically struggling students at an early stage, which reduces the effectiveness of academic interventions.

Another limitation of existing systems is the lack of structured communication between students and faculty. Feedback, assignments, and announcements are often shared through different channels, leading to fragmented communication and difficulty in tracking academic activities. Additionally, administrators have limited tools to monitor institutional performance or evaluate teaching effectiveness based on student outcomes.

Overall, the existing systems provide basic data storage and management functions but lack advanced analytics, centralized access, and intelligent decision-making capabilities. These limitations highlight the need for a more integrated and intelligent academic management platform capable of combining data management, predictive analytics, and communication features to improve educational outcomes.

Limitations of Existing System

Despite improvements in academic management systems, the existing system suffers from the following challenges:

• **Lack of centralized data integration:** Academic data such as attendance, marks, assignments, and feedback are often stored in different systems or maintained manually. This fragmentation makes it difficult to obtain a complete view of a student's academic performance and reduces the efficiency of monitoring and analysis.

• **Limited analytical capabilities:** Traditional systems mainly store academic records without providing predictive insights. Faculty members must manually analyze marks and attendance to identify academically weak students, which can lead to delays in providing timely academic support.

• **Inefficient communication and monitoring:** Communication between faculty and students is often handled through informal platforms such as emails or messaging applications. This lack of structured communication makes it difficult to track academic activities, feedback, and progress systematically.

• **Lack of intelligent decision support:** Many existing academic management systems do not integrate machine learning or predictive analytics. As a result, institutions cannot automatically identify students who may be at risk of poor academic performance, limiting the ability to take proactive measures for academic improvement.

2.4 PROPOSED SYSTEM

The proposed system introduces a centralized web-based platform for managing academic activities and predicting student performance using machine learning techniques. The system integrates academic data such as attendance records, assignment submissions, examination marks, and faculty feedback into a unified database to provide a comprehensive view of student progress. To support predictive analysis, machine learning algorithms such as Naïve Bayes and XGBoost are applied to historical academic data to classify student performance into categories such as Good, Fair, or Poor. The system includes role-based dashboards for administrators, faculty members, and students, enabling efficient management of academic records and communication. Faculty members can upload study materials, assign coursework, record attendance, and provide feedback, while students can access learning resources, view academic reports, and receive machine learning–based performance predictions. The platform also incorporates features such as automated email notifications, structured feedback collection, and faculty performance evaluation based on student results. By combining centralized academic management with predictive analytics, the proposed system improves transparency, enhances decision-making, and enables early identification of students who may require academic support, thereby contributing to improved learning outcomes in educational institutions.

Advantages of the Proposed System:

The proposed system significantly improves upon existing academic management approaches by addressing several key limitations:

• **Centralized Data Management:** The system integrates attendance, marks, assignments, and faculty feedback into a single unified platform, eliminating fragmented data storage and improving transparency in academic monitoring.

• **Machine Learning-Based Prediction:** The use of algorithms such as Naïve Bayes and XGBoost enables accurate prediction of student performance categories, helping faculty identify academically at-risk students at an early stage.

• **Improved Academic Monitoring:** The centralized dashboard allows administrators and faculty members to track student progress, analyze academic trends, and generate performance reports efficiently.

• **Efficient Communication:** The system provides structured communication features such as messaging and automated email notifications, improving interaction between students, faculty, and administrators.

• **Better Decision Support:** Machine learning–based insights help institutions make data-driven academic decisions, enabling timely interventions and improving overall educational outcomes.

2.5 OBJECTIVES

▪ **Develop a Centralized Academic Platform** – Build a web-based system that integrates student and faculty academic activities into a single platform.

▪ **Implement Machine Learning-Based Prediction** – Apply algorithms such as Naïve Bayes and XGBoost to analyze academic data and predict student performance.

▪ **Improve Academic Monitoring Efficiency** – Enable faculty and administrators to track student progress and identify students who require additional academic support.

▪ **Enhance Communication Between Stakeholders** – Provide messaging and notification features that improve interaction between students, faculty, and administrators.

▪ **Support Data-Driven Academic Decisions** – Use predictive analytics to provide actionable insights that help improve learning outcomes.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- Processor : Intel Core i5
- Hard disk : 512 SSD
- RAM : 16GB

2.6.2 SOFTWARE REQUIREMENTS:

Software requirements specify the logical characteristics of each interface and software components of the system. The following are some software requirements:

- **Operating System** : Windows 10
- **Language** : Python
- **Back-End** : Django
- **Database** : MySQL
- **Machine Learning Libraries** : Scikit-learn, XGBoost, NumPy, Pandas
- **Framework** : Django Framework

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, improving collaboration among stakeholders and reducing system complexity.

3.1 PROJECT ARCHITECTURE

This project architecture illustrates the workflow followed in the Centralized Student–Faculty Platform for Real-Time Academic Performance Prediction, starting from data collection to final prediction of student performance.

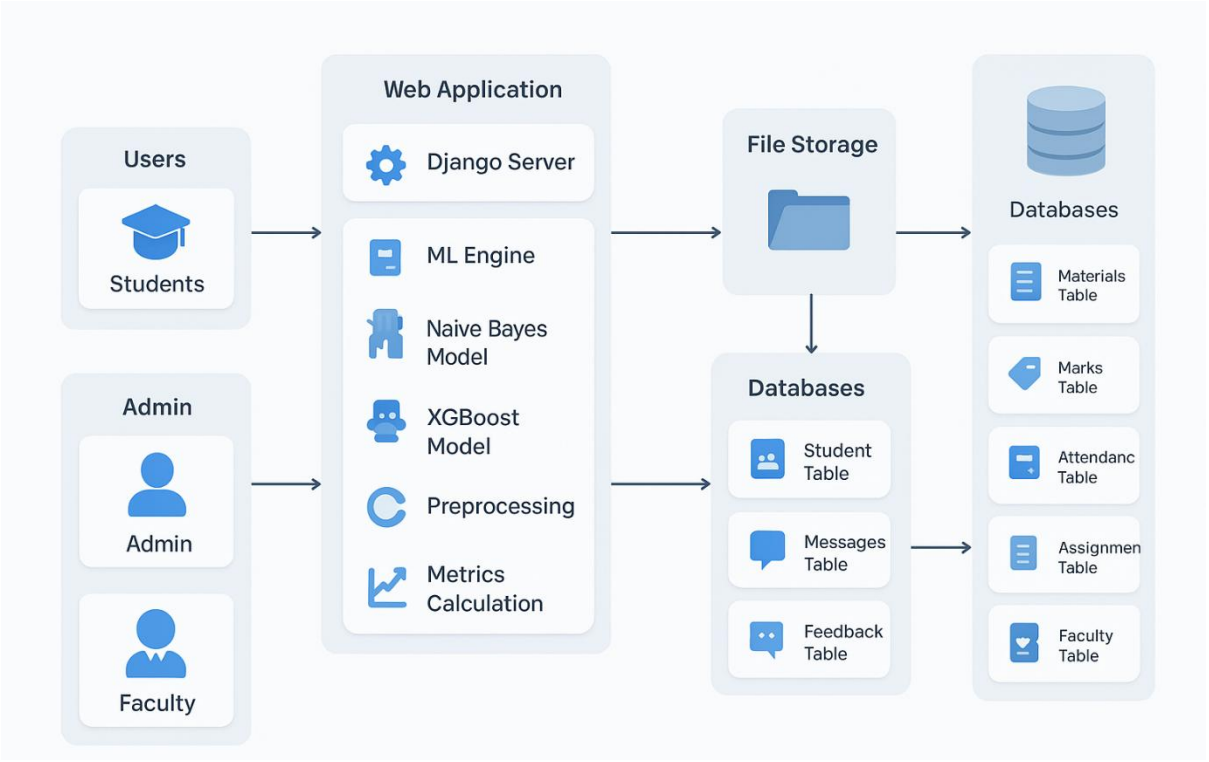


Figure 3.1: Project Architecture of the Centralized Student–Faculty Platform for Real-Time Academic Performance Prediction

3.2 DESCRIPTION

Input Data:

The project uses academic data collected from students and faculty members. The dataset includes information such as attendance records, assignment submissions, examination marks, and course details that represent students' academic performance.

Reading Data:

The collected academic records are retrieved from the database and organized into structured datasets using data handling libraries such as Pandas and NumPy for further analysis.

Data Preprocessing:

The raw data is cleaned and processed by handling missing values, converting data types, and selecting relevant features such as marks, course name, and academic year to prepare the dataset for machine learning algorithms.

Feature Selection:

Important attributes such as total marks, subject scores, and course information are selected to represent student performance effectively and improve prediction accuracy.

Machine Learning Model:

Machine learning algorithms such as Naïve Bayes and XGBoost are used to analyze historical academic data and identify patterns that influence student performance.

Prediction Layer:

The trained models classify student performance into categories such as Good, Fair, or Poor, helping faculty members understand academic progress.

Training and Evaluation:

The models are trained using historical student data and evaluated using performance metrics such as accuracy, precision, recall, and F1-score to measure prediction effectiveness.

Feedback:

The predicted performance results are displayed to faculty members and students through the system dashboard. This feedback helps faculty identify students who require additional academic support and allows students to monitor their progress.

3.3 UNIFIED MODELING LANGUAGE(UML)

Unified Modeling Language (UML) is not just a static set of diagrams; it's a versatile toolset that supports multiple views and perspectives of a system. This versatility allows UML to be applied at different stages of the software development lifecycle, from initial requirements analysis and design to implementation, testing, and maintenance. UML diagrams serve as a common visual language that bridges the gap between technical and non-technical stakeholders. They provide a means for software engineers to communicate complex system designs in a clear and concise manner, facilitating better understanding and collaboration among team members. One of the key benefits of UML is its adaptability to different software development methodologies, such as waterfall, agile, and iterative approaches. UML diagrams can be used to model systems at various levels of abstraction, from high-level architectural views to detailed design specifications. Furthermore, UML is not limited to software systems alone; it can also be applied to model and analyze business processes, organizational structures, and even hardware systems. This flexibility makes UML a valuable tool for a wide range of industries and domains.

GOALS OF UML

1. Provide users a ready-to-use, expressive visual modeling language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development processes.
4. Provide a formal basis for understanding the modeling language.

3.3.1 Class Diagram

The class diagram outlines the core entities in the system such as Student, Faculty, Marks, Assignment, and Attendance. Each class encapsulates relevant attributes and is associated with other classes through well-defined relationships. For instance, a Faculty can upload multiple Assignments, and each Student may have multiple entries in the Marks and Attendance tables. This structure ensures a normalized, relational representation of data, critical for database-backed web applications.

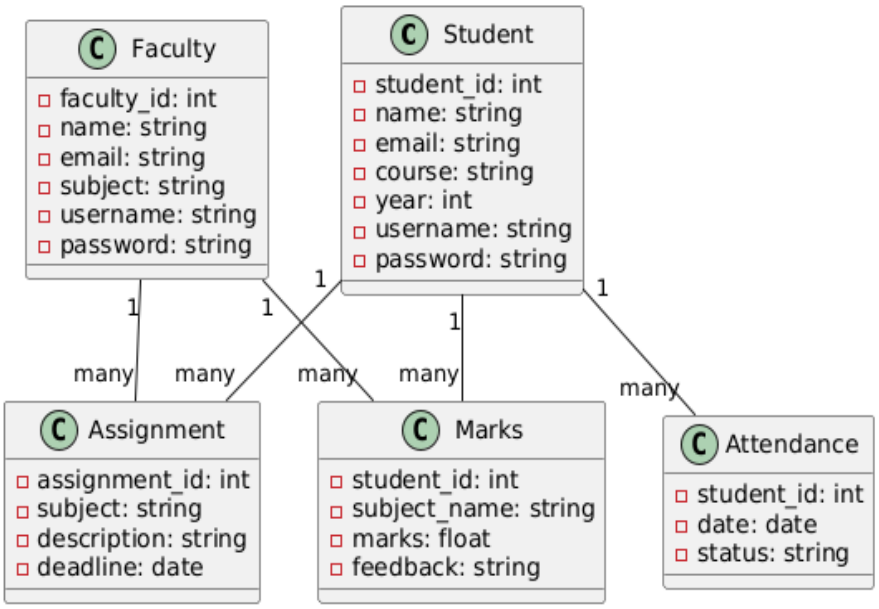


Fig.3.2: Class diagram.

3.3.2 Use Case Diagram

The use case diagram focuses on identifying the different actors in the system — primarily Admin, Faculty, and Student — and mapping them to their respective actions. Admin manages users and views analytics, Faculty handles academic records and communication, and Students can view their progress and feedback. These roles define the boundaries and permissions for each user type, which is essential for building a secure and role-driven application.

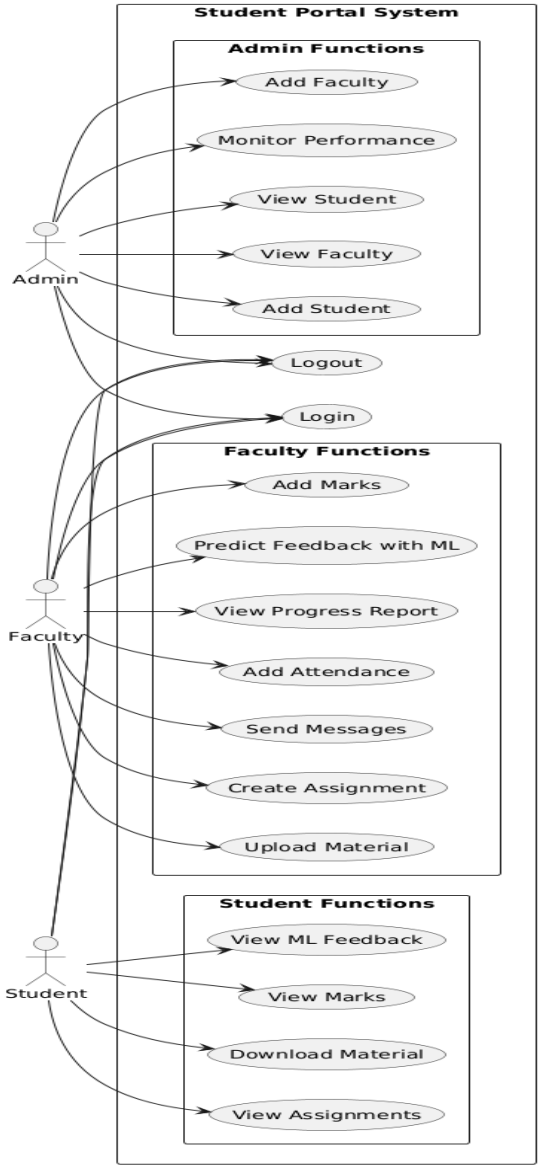


Fig.3.3: Use Case diagram.

3.3.3 Sequence Diagram

The sequence diagram describes the real-time interaction flow for generating ML-based feedback. When a faculty member requests a student's progress report, the system fetches relevant data from the database, computes the average, and sends it to the ML model. The model then predicts the performance level and returns feedback, which is finally rendered to the user. This sequence ensures a smooth, responsive experience where machine learning is seamlessly integrated into the traditional educational workflow.

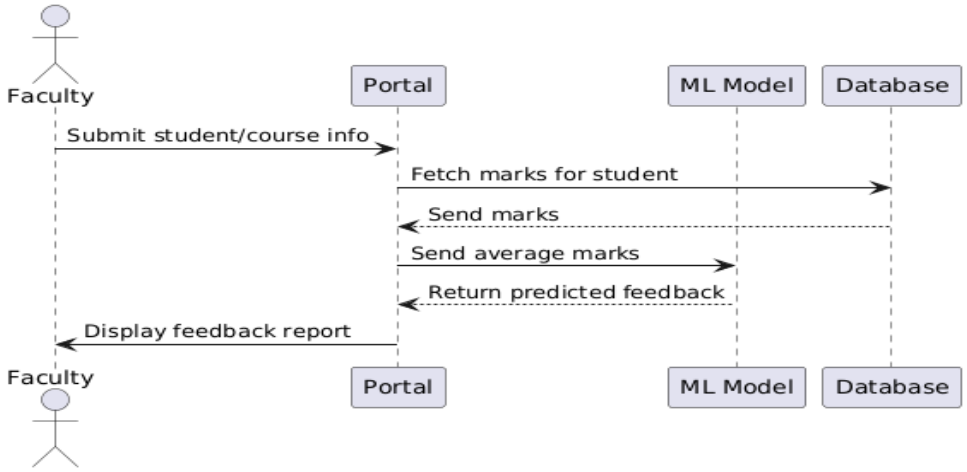


Fig.3.4: Sequence diagram.

3.3.4 Component Diagram

The component diagram breaks down the system into modular, interchangeable parts — UI, controllers (Django views), database, and the ML feedback module. Each component handles a specific responsibility and communicates with others through defined interfaces. This modularity allows easier debugging, testing, and future scalability.

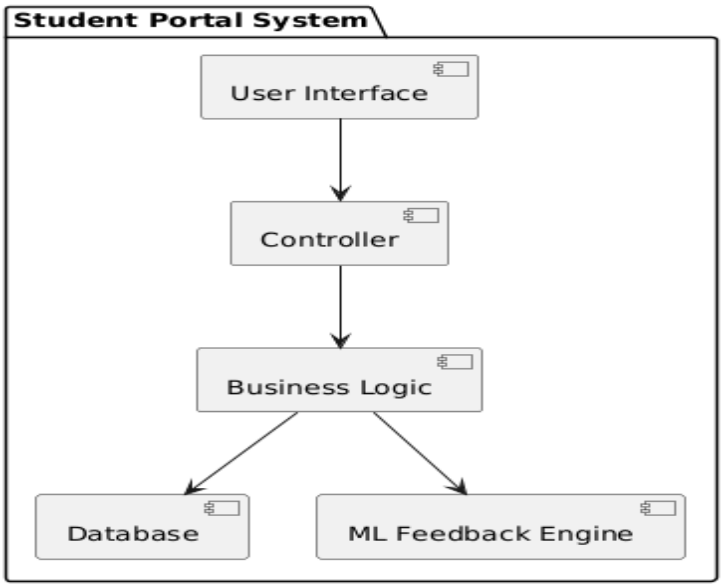


Fig.3.5: Component diagram.

3.3.5 Deployment Diagram

The deployment diagram represents the system's physical setup. It comprises multiple nodes such as a Web Server (Django application), a Database Server (MySQL), and an ML Server (hosting the model). These are connected over a local network or internet and demonstrate how the software system will run in a real-world infrastructure.

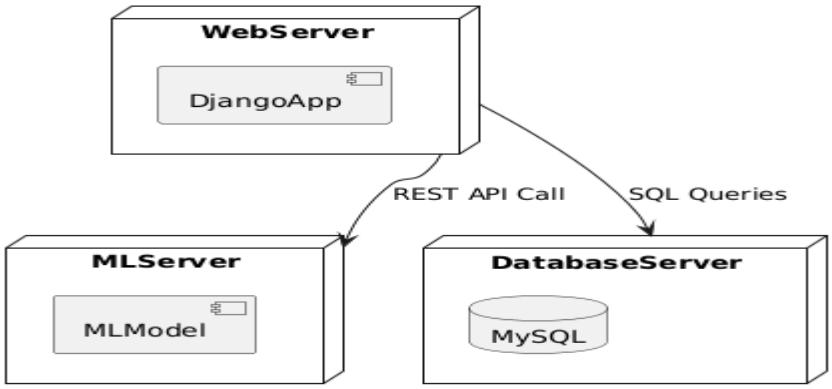


Fig.3.6: Deployment diagram.

3.3.6 Activity Diagram

The activity diagram visually describes the flow of actions starting from faculty login to viewing the student's performance report. It includes conditional decisions such as validating input, processing data, calling ML models, and displaying output. This helps in understanding the operational flow of the system and identifying any bottlenecks or redundant processes.

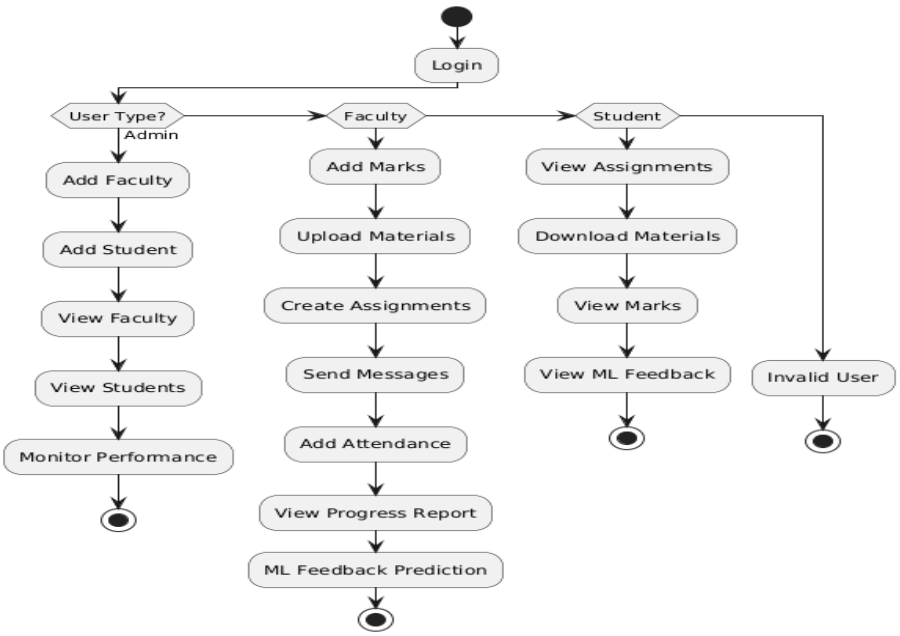


Fig.3.7: Activity diagram.

3.3.7 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows within a system, showing the interaction between processes, data stores, and external entities. It is an important tool used in system analysis and design because it helps to visualize how information moves through the system and how different components interact with each other. By representing the flow of data clearly, a DFD enables developers, analysts, and stakeholders to understand system functionality and identify possible improvements.

A Data Flow Diagram consists of four main components:

• **External Entities:** These represent sources or destinations of data outside the system, such as students, faculty, or administrators interacting with the platform.

• **Processes:** These represent operations performed on data, such as processing student marks, generating performance predictions, or updating attendance records.

• **Data Flows:** These illustrate how data moves between processes, external entities, and data stores within the system.

• **Data Stores:** These indicate where the data is stored in the system, such as databases containing student information, academic records, or feedback details.

These elements are represented using standard symbols. Rectangles denote external entities, circles represent processes, arrows show the direction of data flow, and open-ended rectangles indicate data stores. Using these standardized symbols ensures clarity and consistency while designing system models.

Benefits:

The graphical nature of DFDs allows both technical and non-technical stakeholders to easily understand how the system works. It helps identify inefficiencies, redundant processes, and potential security risks in the data flow. In addition, DFDs assist developers in improving system structure and ensure efficient handling of data throughout the application.

Applications:

Data Flow Diagrams are widely used in software development, information system design, and business process analysis. They are useful for modeling data movement, documenting system architecture, and identifying areas where automation or optimization can improve system performance.

In conclusion, a Data Flow Diagram is a crucial modeling tool that provides a clear visual understanding of how data moves through a system. It simplifies complex processes and supports effective communication during the development and implementation of software systems.

Levels of DFD:

DFDs are organized into multiple levels to provide increasing detail about the system processes.

• **Level 0 (Context Diagram):**

This level represents the overall system as a single process and shows the interaction between the system and external entities such as students, faculty, and administrators.

• **Level 1:**

This level expands the main process into multiple sub-processes, such as student data management, faculty interaction, attendance management, and performance prediction modules.

• **Level 2+:**

Further levels provide a more detailed view of specific processes within the system, breaking them down into smaller operations to better understand complex workflows and system functionality.

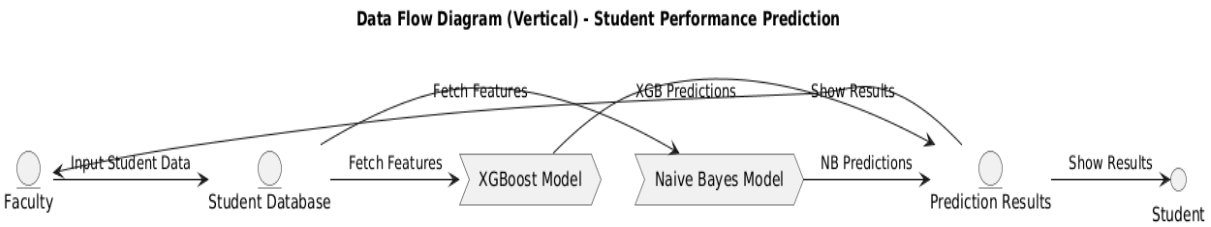


Figure 3.8 : Dataflow Diagram – Student Performance Prediction

4. IMPLEMENTATION

4 IMPLEMENTATION

The implementation phase of the project focuses on converting the designed system into a functional platform by executing the planned modules and integrating machine learning techniques with the web application. It involves coordinating different components such as data collection, preprocessing, database integration, machine learning model training, and system deployment. Proper implementation ensures that the centralized student–faculty platform functions efficiently, allowing real-time academic performance prediction and effective communication among stakeholders. The system integrates web technologies with predictive analytics to deliver accurate insights and improve academic decision-making within the specified time and resource constraints.

4.1 ALGORITHMS USED

They require large datasets for training and can be computationally expensive when handling high-resolution videos.

Machine Learning Models for Academic Performance Prediction

Machine learning algorithms are widely used for predicting academic performance because they can analyze large amounts of educational data and identify hidden patterns. In this project, algorithms such as **Naive Bayes and XGBoost** are employed to classify student performance levels based on historical academic records including marks, attendance, and assignments. These models process structured datasets to predict whether a student's performance falls into categories such as Good, Fair, or Poor.

The system collects academic data from faculty inputs and student activities, preprocesses the data, and feeds it into the machine learning models. The models learn from historical data and generate predictions that help faculty identify students who may require additional support. This approach enables institutions to make proactive academic decisions and improve overall learning outcomes.

Advantages of Machine Learning Models

• Capable of analyzing large volumes of academic data efficiently.
• Provide predictive insights that help faculty identify at-risk students early.
• Improve decision-making by offering data-driven recommendations.
• Reduce manual analysis of academic records.

Disadvantages of Machine Learning Models

• Require properly preprocessed and structured datasets.
• Prediction accuracy depends on the quality and quantity of training data.
• Some models may require parameter tuning and computational resources.

Naive Bayes Classifier

The Naive Bayes classifier is a probabilistic machine learning model based on Bayes' Theorem. It assumes independence between features, which makes it simple and efficient, especially for classification tasks involving textual or categorical data. Despite its simplicity, Naive Bayes often performs surprisingly well in real-world scenarios, especially when the dataset is not very large.

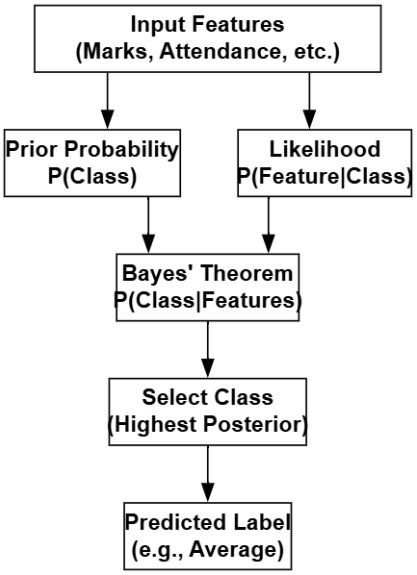


Fig. 4.1: Internal architecture of Naive Bayes classifier for student performance prediction.

Internal Working

- **Step 1 Prior Probability Calculation**: Calculates the prior probability of each class (e.g., Good, Average, Poor) based on training data.
- **Step 2 Likelihood Estimation**: Computes the likelihood of each feature given a class, assuming features are conditionally independent.
- **Step 3 Posterior Probability**: Applies Bayes' Theorem:

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

Where for a classification problem:

- P(C|X): The Posterior probability what we want to find. The probability that the sample {X} belongs to class C.
- P(C): The Prior probability. The frequency of class C in the training data.
- P(X|C): The Likelihood. The probability of observing features {X} given that the class is C.
- P(X): The Evidence. The probability of observing the features {X}, which acts as a scaling factor (it's often constant across all classes, so it can be ignored when just comparing probabilities).

- **Step 4 Prediction**: The class with the highest posterior probability is chosen as the predicted class.
- **Advantages**: Fast, simple, efficient for small to medium-sized datasets. Handles categorical input well.

XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a powerful ensemble machine learning model based on decision trees. It builds multiple trees in sequence, where each tree learns from the errors of its predecessors. XGBoost is known for its speed, regularization techniques, and high performance on structured/tabular data.

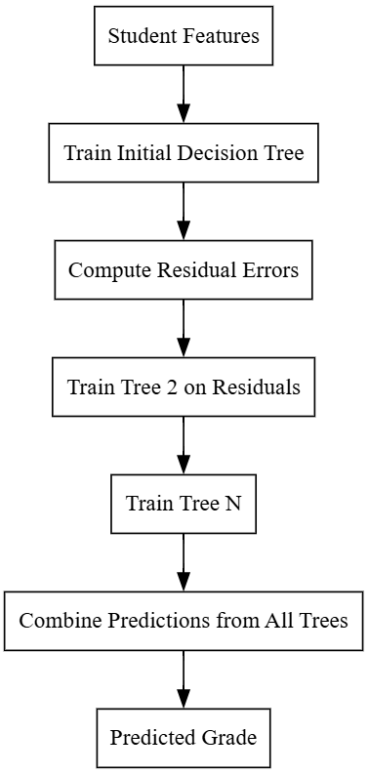


Fig. 4.2: Flow diagram of XGBoost classifier training process.

Internal Working

- **Step 1: Initialization**
  - Starts with an initial prediction (typically the mean of target values for regression or log-odds for classification).
- **Step 2: Gradient Calculation**
  - Calculates gradients (errors or residuals) for each prediction compared to actual values.
- **Step 3: Tree Building**
  - Builds a decision tree to predict the gradients (residuals). The split is determined by minimizing a loss function.
- **Step 4: Update Model**
  - Combines the new tree with previous trees using a learning rate to improve prediction.

- **Step 5: Regularization**
  - Applies L1/L2 regularization to prevent overfitting by penalizing complex trees.
- **Step 6: Repeat**
  - Continues to add new trees until a stopping criterion is met (number of trees, accuracy, or convergence).

Advantages

- Handles missing values automatically.
- Highly scalable and accurate.
- Supports parallel processing and cross-validation.

Now-a-days educational institutions generate a large amount of academic data related to students such as attendance records, assignment submissions, internal assessment marks, and examination results. Managing and analyzing this information manually is a time-consuming task and often leads to errors in academic evaluation. Traditional academic management systems mainly focus on storing records rather than analyzing them effectively. In many institutions, student information is maintained in different files or independent systems, which makes it difficult for faculty members and administrators to track student progress efficiently.

To overcome these limitations, a centralized system is required that can manage academic data and provide meaningful insights about student performance. In this project, a centralized student–faculty platform is developed to collect, manage, and analyze academic data in a structured manner. Machine learning techniques are used to analyze student records and predict academic performance based on historical data such as marks, attendance, and assignment scores.

The system allows faculty members to monitor student progress and identify students who may require additional academic support. By integrating machine learning algorithms with a centralized platform, the proposed system improves decision-making and helps educational institutions enhance the overall learning experience for students.

To implement the centralized student–faculty platform, the following modules are developed:

1. **Student and Faculty Registration Module:** Allows administrators to register students and faculty members in the system.
2. **Academic Data Management Module:** Enables faculty to upload marks, assignments, and attendance details.
3. **Data Preprocessing Module:** Cleans and structures academic data for machine learning analysis.
4. **Machine Learning Prediction Module:** Applies Naive Bayes and XGBoost algorithms to predict student performance.
5. **Dashboard and Visualization Module:** Displays performance reports, predictions, and analytics to users.
6. **Communication Module:** Enables messaging and feedback between faculty and students.
7. **Performance Report Generation Module:** Generates academic progress reports and predictive insights.

4.2 SAMPLE CODE

```
from django.shortcuts import render

from django.template import RequestContext

from django.contrib import messages

from django.http import HttpResponse

import os

import pickle

import pymysql

from django.core.files.storage import FileSystemStorage

from datetime import datetime

import io

import base64

import numpy as np

import matplotlib.pyplot as plt

import smtplib

import pandas as pd

from sklearn.preprocessing import LabelEncoder

from sklearn.preprocessing import MinMaxScaler

from sklearn.model_selection import train_test_split

from sklearn.metrics import precision_score

from sklearn.metrics import recall_score

from sklearn.metrics import f1_score

import seaborn as sns
```

```
from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score

from sklearn.metrics import confusion_matrix

from xgboost import XGBClassifier

global uname

dataset = pd.read_csv("model/dataset.csv")

labels = np.unique(dataset['FinalResult'])

le = LabelEncoder()

dataset['FinalResult'] = pd.Series(

le.fit_transform(dataset['FinalResult'].astype(str))

)

Y = dataset['FinalResult'].ravel()

dataset = dataset.values

X = dataset[:,0:dataset.shape[1]-1]

sc = MinMaxScaler()

X = sc.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(

X, Y, test_size=0.2

)

accuracy=[]

precision=[]

recall=[]
```

```
fscore=[]

def calculateMetrics(algorithm, y_test, predict):

 a = accuracy_score(y_test,predict)*100

    p = precision_score(

    y_test,predict,average='macro'

    ) * 100

    r = recall_score(

    y_test,predict,average='macro'

    ) * 100

    f = f1_score(

    y_test,predict,average='macro'

    ) * 100

    accuracy.append(a)

    precision.append(p)

    recall.append(r)

    fscore.append(f)

nb_cls = GaussianNB()

nb_cls.fit(X_train, y_train)

predict = nb_cls.predict(X_test)

calculateMetrics("Naive Bayes",y_test,predict)

xg_cls = XGBClassifier()

xg_cls.fit(X_train, y_train)

predict = xg_cls.predict(X_test)

calculateMetrics("XGBoost",y_test,predict)
```

```
def runML(marks):
    data=[]
data.append([marks])
    data=np.asarray(data)
    data=sc.transform(data)
    predict=xg_cls.predict(data)[0]
    return labels[predict]
def ViewMarks(request):
    if request.method=='GET':
        output="<table border=1>"
        output+="<tr><th>Subject</th><th>Marks</th></tr>"
        con=pymysql.connect(
        host='127.0.0.1',
        user='root',
        password='root',
        database='studentportal'
        )
        with con:
            cur=con.cursor()
            cur.execute(
            "select subject_name,subject_marks from marks where student_name='"+uname+"'"
            )
            rows=cur.fetchall()
  total=0
```

```
        count=0
 for row in rows:
            output+="<tr><td>"+row[0]+"</td><td>"+str(row[1])+"</td></tr>"
            total+=row[1]
            count+=1
      avg=total/count
      prediction=runML(avg)
      output+="<tr><td>Total</td><td>"+str(total)+"</td></tr>"
      output+="<tr><td>Average</td><td>"+str(avg)+"</td></tr>"
      output+="<tr><td>Prediction</td><td>"+prediction+"</td></tr>"
      output+="</table>"
      context={'data':output}
      return render(request,'StudentScreen.html',context)
def FacultySuggestion(request):
   if request.method=='GET':
      output=""
      output+="<table border=1 align=center width=100%>"
      output+="<tr><th>Faculty Name</th><th>Performance %</th></tr>"
      con=pymysql.connect(
      host='127.0.0.1',
      user='root',
      password='root',
      database='studentportal'
      )
   with con:
```

```
        cur=con.cursor()
        cur.execute(
        "select faculty_name,avg(subject_marks) from marks group by faculty_name"
        )
        rows=cur.fetchall()
 for row in rows:
            output+="<tr>"
            output+="<td>"+row[0]+"</td>"
            output+="<td>"+str(row[1])+"</td>"
            output+="</tr>"
      output+="</table>"
      context={'data':output}
      return render(request,'FacultySuggestion.html',context)
def SubmitFeedback(request):
   if request.method=='POST':
      feedback=request.POST.get('t2',False)
      rating=request.POST.get('t3',False)
      now=datetime.now()
 current_datetime=str(now.strftime("%Y-%m-%d %H:%M:%S"))
      current_datetime=current_datetime.split(" ")[0].strip()
      db_connection=pymysql.connect(
      host='127.0.0.1',
      user='root',
  password='root',
```

```
        database='studentportal'
        )
        db_cursor=db_connection.cursor()
        student_sql_query=(
        "INSERT INTO feedback VALUES('"
        +uname+"','faculty','"
        +feedback+"','"+rating+"','"
        +current_datetime+"')"
        )
        db_cursor.execute(student_sql_query)
        db_connection.commit()
        status="error in submitting feedback"
        if db_cursor.rowcount==1:
            status="Your feedback successfully submitted"
        context={'data':status}
        return render(request,'StudentScreen.html',context)
def StudentLoginAction(request):
    if request.method=='POST':
        username=request.POST.get('username')
        password=request.POST.get('password')
        con=pymysql.connect(
        host='127.0.0.1',
        user='root',
  password='root',
```

```
    database='studentportal'
    )
    with con:
        cur=con.cursor()
        cur.execute(
        "select username,password from student where username='"+username+"' and
password='"+password+"'"
        )
        rows=cur.fetchall()
        if len(rows)>0:
            global uname
            uname=username
            context={'data':'Welcome '+username}
            return render(request,'StudentScreen.html',context)
        else:
            context={'data':'Login failed'}
            return render(request,'StudentLogin.html',context)
def AdminLoginAction(request):
  if request.method=='POST':
    username=request.POST.get('username')
    password=request.POST.get('password')
    if username=='admin' and password=='admin':
        context={'data':'Welcome Admin'}
  return render(request,'AdminScreen.html',context)
```

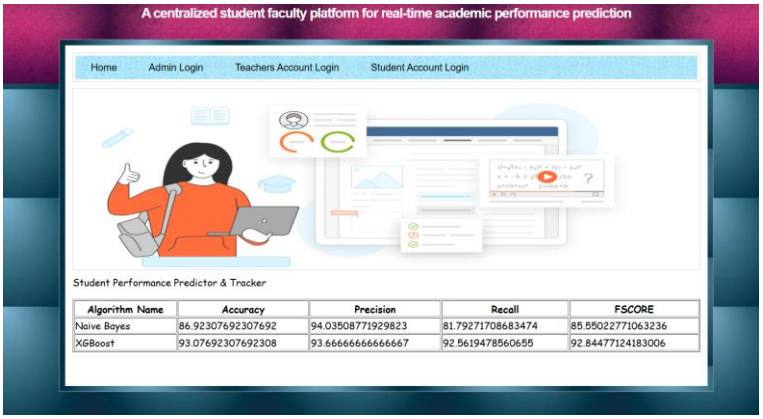
```
        else:
            context={'data':'Invalid Login'}
            return render(request,'AdminLogin.html',context)
def SendMessage(request):
    if request.method=='POST':
        sender=request.POST.get('sender')
        receiver=request.POST.get('receiver')
        message=request.POST.get('message')
        con=pymysql.connect(
        host='127.0.0.1',
        user='root',
        password='root',
        database='studentportal'
        )
        with con:
            cur=con.cursor()
            sql=(
            "INSERT INTO messages VALUES('"
            +sender+"','"+receiver+"','"+message+"')"
            )
            cur.execute(sql)
            con.commit()
        context={'data':'Message Sent Successfully'}
                                return render(request,'SendMessage.html',cont
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

The system produced accurate and meaningful results in managing academic data as well as predicting student performance using machine learning. Blow screens are able to view the marks, attendance, assignments, study materials, and academic updates through their personalized dashboard, while faculty and administrators efficiently performed academic management tasks such as entering marks, uploading material, and reviewing student progress.



Algorithm Name	Accuracy	Precision	Recall	FSCORE
Naive Bayes	86.92307692307692	94.03508771929823	81.79271708683474	85.55022771063236
XGBoost	93.07692307692308	93.66666666666667	92.56194785606555	92.84477124183006

Fig. 5.1: Interface overview.

Fig. 5.1 displays the main interface of the system, providing an organized dashboard for both teachers and students. It includes navigation options for uploading data, creating assignments, managing attendance, and accessing learning materials. The interface ensures easy access to all core functionalities of the project.

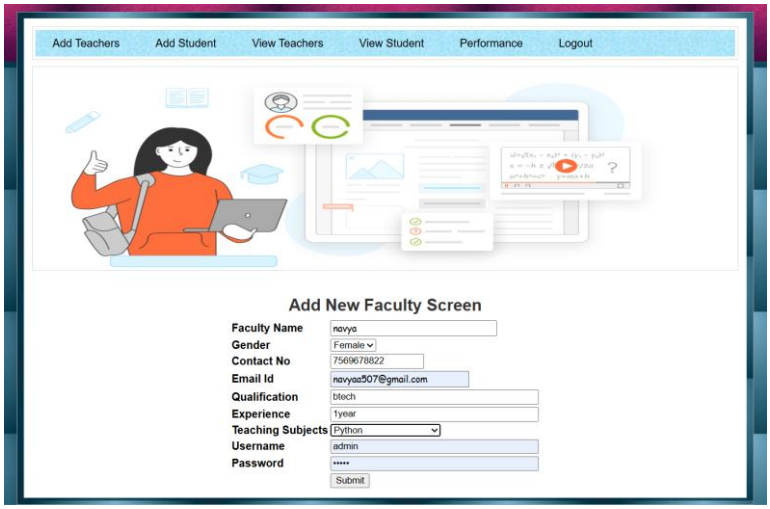


Fig. 5.2: Uploading teacher details

Fig. 5.2 demonstrates the functionality where administrator or authorized personnel upload teacher information into the system. The uploaded data includes teacher names, subjects assigned, and contact information. This process ensures the system maintains a structured record of teaching staff for academic management.

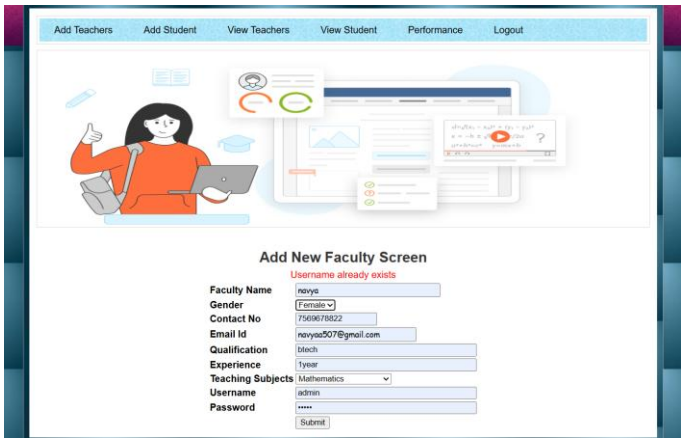


Fig. 5.3: Uploading student details.

Fig. 5.3 illustrates the process of adding student information into the system. The data includes student names, roll numbers, class, and relevant academic details. Uploading these details enables the system to track individual student performance and link them to their respective courses and teachers.

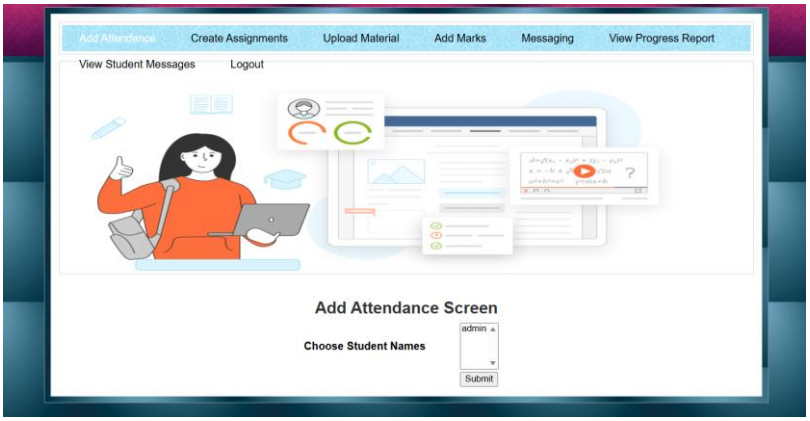


Fig. 5.4: Teacher uploading student attendance.

Fig.5.4 shows the teacher marking or uploading attendance for students. Attendance data is recorded and associated with specific dates and subjects. This feature ensures accurate tracking of student presence and supports performance analysis.

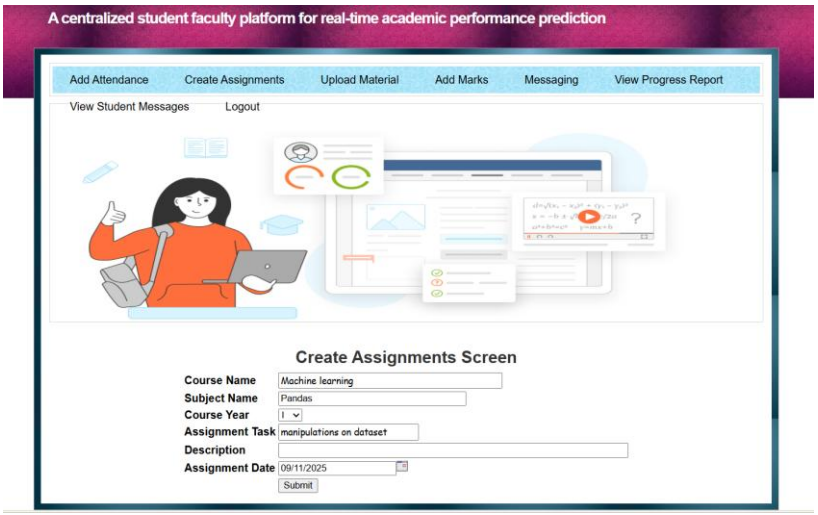


Fig.5.5: Teacher creating assignments.

Fig.5.5 presents the process where teachers create assignments for students. Teachers input assignment details, including title, description, and submission deadlines. The system stores these assignments, making them accessible for students to review and submit.

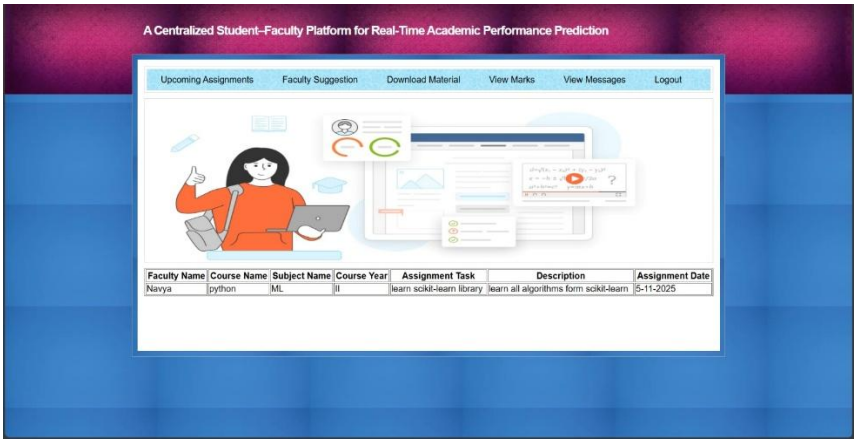


Fig.5.6: Student checking assignments.

Fig.5.6 illustrates the student interface for viewing assignments assigned by teachers. Students can access assignment instructions, deadlines, and submission status. This feature facilitates timely completion and submission of academic tasks.

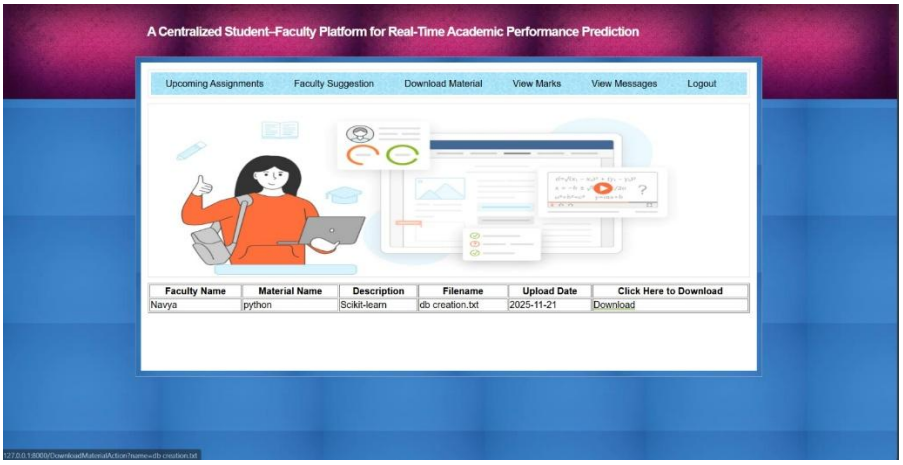


Fig. 5.7: Student downloading learning materials.

Fig. 5.7 displays the functionality where students download uploaded study materials. Materials such as notes, presentations, and reference documents are organized by subject. This feature ensures students have easy access to resources required for learning and assignment completion.

6. VALIDATION

7. VALIDATION

The validation of this project focuses on verifying the accuracy, reliability, and effectiveness of the centralized student–faculty platform designed for real-time academic performance prediction. The validation process involves systematic testing of the system modules, dataset processing, and machine learning model evaluation. Through structured validation methods, the system ensures that academic data such as student marks, attendance, assignments, and feedback are processed correctly and predictions are generated accurately.

The validation procedure includes multiple stages such as data validation, system functionality testing, and predictive model performance evaluation. During validation, the system checks whether the academic data entered by faculty members is correctly stored in the centralized database and used effectively for generating performance predictions. By implementing this structured validation process, the system ensures reliable prediction results and accurate academic insights for both faculty and students.

i. INTRODUCTION

Initially, the dataset containing student academic records such as marks, attendance, and course information is divided into training and testing datasets. Typically, around 70–80% of the dataset is used for training the machine learning models while the remaining data is reserved for testing. The training dataset is used to train predictive algorithms such as Naïve Bayes and XGBoost, whereas the testing dataset evaluates the performance of these models on unseen student data.

To ensure reliability and accuracy, model validation techniques such as cross-validation are used. These techniques help reduce overfitting and ensure that the model generalizes well to new academic data. Through this process, the predictive models learn patterns from historical student performance data and generate predictions such as Good, Average, or Poor performance categories.

The performance of the prediction system is measured using evaluation metrics including accuracy, precision, recall, and F1-score. These metrics help determine how effectively the system predicts student performance. Additionally, confusion matrix analysis provides detailed insight into correct and incorrect predictions, allowing improvements to the prediction model.

Finally, system validation is performed by testing the platform in real-time scenarios where faculty upload marks, students access their performance reports, and the machine learning module generates predictive feedback. This testing confirms that the platform operates efficiently in real academic environments. The validation process ensures that the centralized student–faculty platform is reliable, scalable, and capable of delivering accurate academic performance predictions.

6.2 TEST CASES

TABLE 6.2.1 Comparative Analysis

Algorithms Name	Accuracy	Precision	Recall	F1-Score
Naive Bayes Classifier	86.92	94.03	81.79	80.55
XGBoost Classifier	93.07	93.66	93.56	92.84

Table 1: Performance comparison for the Naive Bayes Classifier and XGBoost algorithms.

The Table 1 presents a performance comparison between the Naive Bayes Classifier and the XGBoost algorithm. The Naive Bayes Classifier achieved an accuracy of 86.92%, with precision, recall, and F1-score values of 94.03%, 81.79%, and 80.55%, respectively. In contrast, the XGBoost Classifier outperformed Naive Bayes across all metrics, achieving an accuracy of 93.07%, precision of 93.66%, recall of 93.56%, and an F1-score of 92.84%. This comparison highlights the stronger predictive capability of XGBoost in handling the dataset used for this project. The results demonstrate that XGBoost provides more balanced performance between precision and recall, ensuring higher reliability in classification tasks. Overall, XGBoost offers a significant improvement in accuracy and effectiveness compared to the Naive Bayes approach.

TABLE 6.2.2 Performance metrics of Naive Bayes Classifier.

Metric	Value
Accuracy	86.92
Precision	94.03
Recall	81.79
F1-Score	80.55

Table 2 shows the performance metrics of the Naive Bayes Classifier on the dataset. The model achieved an accuracy of 86.92%, indicating a strong overall correctness in predictions. Precision reached 94.03%, reflecting a high proportion of correctly identified positive instances. Recall was 81.79%, showing the model's ability to identify relevant cases, while the F1-score of 80.55% represents a balanced measure of precision and recall.

TABLE 6.2.3 Performance metrics of XGBoost Classifier.

Metric	Value
Accuracy	93.07
Precision	93.66
Recall	93.56
F1-Score	92.84

Table 3 presents the performance metrics of the XGBoost Classifier on the dataset. The model achieved an accuracy of 93.07%, demonstrating excellent overall prediction capability. Precision was 93.66%, indicating a high rate of correctly predicted positive instances. Recall reached 93.56%, reflecting the model's effectiveness in identifying relevant cases. The F1-score of 92.84% highlights a strong balance between precision and recall, confirming the robustness of the XGBoost model.

7. CONCLUSION & FUTURE ASPECTS

8. CONCLUSION & FUTURE ASPECTS

In conclusion, the project **"A Centralized Student–Faculty Platform for Real-Time Academic Performance Prediction"** has successfully achieved its objectives by developing an academic management system that integrates centralized data handling with machine learning–based prediction. The implementation phases were carefully planned and executed, resulting in a platform that allows administrators, faculty members, and students to interact within a single academic environment. The system improves the monitoring of academic activities such as marks entry, assignment management, and communication between faculty and students.

Looking ahead, the future aspects of the project hold strong potential for further development. Future improvements will focus on expanding system capabilities, integrating advanced machine learning techniques, and improving scalability. These enhancements will strengthen the existing framework and open new opportunities for innovation in academic performance monitoring. This approach will ensure that the platform remains useful and impactful for modern educational institutions.

7.1 PROJECT CONCLUSION

In this research, a **Centralized Student–Faculty Platform for Real-Time Academic Performance Prediction** was successfully developed. The study demonstrated the effectiveness of integrating machine learning techniques with centralized academic data management to monitor and predict student performance. The proposed system utilized algorithms such as **Naïve Bayes and XGBoost** to analyze student academic data including marks, attendance, and assignments.

The models were evaluated using performance metrics such as **accuracy, precision, recall, and F1-score** to measure prediction performance. Among the algorithms used, the **XGBoost classifier achieved better prediction accuracy compared to the Naïve Bayes model**, showing its effectiveness for analyzing academic data.

The improvement in prediction capability is due to the ability of machine learning algorithms to identify patterns in historical academic records.

The system also provides a **web-based interface using the Django framework**, allowing faculty members to upload marks and assignments while students can view their academic performance and prediction results.Overall, the proposed system demonstrates strong potential for improving academic monitoring and supporting data-driven decision making in educational institutions.

7.2 FUTURE ASPECTS

Although the proposed system performed effectively, there are several opportunities for future improvement. The dataset used for training the machine learning models can be expanded with more academic records from different departments and institutions to improve prediction accuracy.

Future work can also focus on integrating additional machine learning techniques such as **Artificial Neural Networks and Random Forest algorithms** to enhance prediction performance. Including more academic features such as attendance patterns and assignment completion data can further improve the system's analytical capabilities.

Another enhancement is deploying the system on **cloud platforms and mobile applications**, allowing students and faculty to access the system easily from multiple devices.

The system can also be integrated with existing **Learning Management Systems (LMS)** to automatically collect academic data and reduce manual work. By incorporating these improvements, the system can evolve into a more scalable and intelligent academic performance monitoring platform for modern educational institutions.

8.BIBLIOGRAPHY

8. BIBLIOGRAPHY

8.1 REFERENCES

[1] Kezar, A.; Kitchen, J.A.; Estes, H.; Hallett, R.; Perez, R. Tailoring programs to best support low-income, first-generation, and racially minoritized college student success. *J. Coll. Stud. Retent. Res. Theory Pract.* **2023**, *25*, 126–152.

[2] Wilmer, E. Student support services for the underprepared student. *Inq. J. Va. Community Coll.* **2008**, *13*, 4–19.

[3] Andrews, R.J.; Imberman, S.A.; Lovenheim, M.F. Recruiting and supporting low-income, high-achieving students at flagship universities. *Econ. Educ. Rev.* **2020**, *74*, 101923.

[4] Hoyt, J.E. Student connections: The critical role of student affairs and academic support services in retention efforts. *J. Coll. Stud. Retent. Res. Theory Pract.* **2023**, *25*, 480–491.

[5] Kuh, G.D. Student success. In *Student Services: A Handbook for the Profession*, 5th ed.; Schuh, J.H., Jones, S.R., Harper, S.R., Eds.; Jossey-Bass: Hoboken, NJ, USA, 2011; pp. 257–270.

[6] Alruwais, N.; Zakariah, M. Evaluating Student Knowledge Assessment Using Machine Learning Techniques. *Sustainability* **2023**, *15*, 6229. https://doi.org/10.3390/su15076229.

[7] Oliveira, P.F.; Matos, P. Introducing a Chatbot to the Web Portal of a Higher Education Institution to Enhance Student Interaction. *Eng. Proc.* **2023**, *56*, 128. https://doi.org/10.3390/ASEC2023-16621.

[8] Imundo, M.N.; Goldshtein, M.; Watanabe, M.; Gong, J.; Crosby, D.N.; Roscoe, R.D.; Arner, T.; McNamara, D.S. Awareness to Action: Student Knowledge of and Responses to an Early Alert System. *Appl. Sci.* **2025**, *15*, 6316. https://doi.org/10.3390/app15116316.

[9] Guanin-Fajardo, J.H.; Guaña-Moya, J.; Casillas, J. Predicting Academic Success of College Students Using Machine Learning Techniques. *Data* **2024**, *9*, 60. https://doi.org/10.3390/data9040060.

[10] Syed Mustapha, S.M.F.D. Predictive Analysis of Students' Learning Performance Using Data Mining Techniques: A Comparative Study of Feature Selection Methods. *Appl. Syst. Innov.* **2023**, *6*, 86. https://doi.org/10.3390/asi6050086.

[11] Zou, W.; Zhong, W.; Du, J.; Yuan, L. Prediction of Student Academic Performance Utilizing a Multi-Model Fusion Approach in the Realm of Machine Learning. *Appl. Sci.* **2025**, *15*, 3550. https://doi.org/10.3390/app15073550.

[12] Luo, Z.; Mai, J.; Feng, C.; Kong, D.; Liu, J.; Ding, Y.; Qi, B.; Zhu, Z. A Method for Prediction and Analysis of Student Performance That Combines Multi-Dimensional Features of Time and Space. *Mathematics* **2024**, *12*, 3597. https://doi.org/10.3390/math12223597.

[13] Chen, S.; Ding, Y. A Machine Learning Approach to Predicting Academic Performance in Pennsylvania's Schools. *Soc. Sci.* **2023**, *12*, 118. https://doi.org/10.3390/socsci12030118.

[14] Alboaneen, D.; Almelihi, M.; Alsubaie, R.; Alghamdi, R.; Alshehri, L.; Alharthi, R. Development of a Web-Based Prediction System for Students' Academic Performance. *Data* **2022**, *7*, 21. https://doi.org/10.3390/data7020021.

[15] Angeioplastis, A.; Aliprantis, J.; Konstantakis, M.; Tsimpiris, A. Predicting Student Performance and Enhancing Learning Outcomes: A Data-Driven Approach Using Educational Data Mining Techniques. *Computers* **2025**, *14*, 83. https://doi.org/10.3390/computers14030083.

[16] Zhao, L.; Ren, J.; Zhang, L.; Zhao, H. Quantitative Analysis and Prediction of Academic Performance of Students Using Machine Learning. *Sustainability* **2023**, *15*, 12531. https://doi.org/10.3390/su151612531.

8.2 GITHUB LINK

https://github.com/mashapathrianjali03/Major_Project_Batch-A23